

Thank you for purchasing a FANUC Robot

READ THIS FIRST

Use this guide to unpack, install, turn on, and jog your new FANUC robot.

The appropriate level of safety for your application and installation can best be determined by safety system professionals. A qualified electrician should apply power. FANUC recommends that each customer consult with such professionals in order to provide a safe application, and take the necessary FANUC training course to operate the robot safely. Refer to your FANUC documentation provided with your robot for safety procedures.

Required Tools

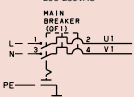
- ◆ Straight-head screwdriver
- ◆ Torque wrench
- ◆ Cross-head screwdriver
- ◆ Metric hex-head wrenches

Required Items

- ◆ Properly rated transport equipment as specified
- ◆ Customer-supplied properly rated power source as specified on the Controller Power Source Label
- ◆ Customer-supplied properly rated power input cable; conductor size of AWG14 to AWG10; terminal size is M5
- ◆ Multimeter
- ◆ Rigid platform to hold robot
- ◆ 4-M10 x 35 mm bolts
- ◆ Locite #243
- ◆ 4-M10 flat washers

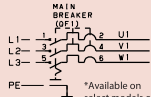
Single Phase

100-120VAC (with transformer option)
200-230VAC



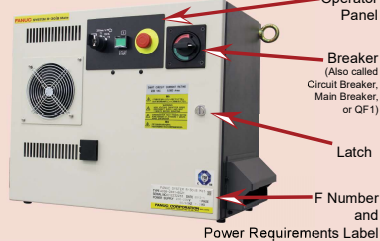
Three Phase*

200-230VAC

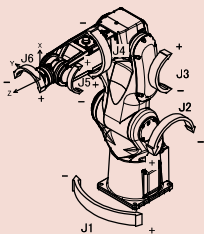


*Available on select models only.

Controller



Robot (mechanical unit) with joints indicated



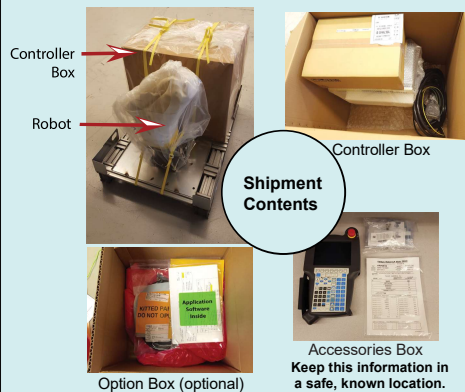
AM 50iD/7L 6-Axis Robot

All axes are 0° in this position

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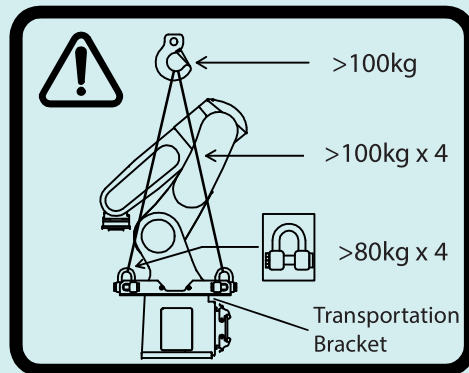
A. Unpack and Intall your Robot



1. Open all of the boxes included in your shipment.
2. Remove the plastic covering from the robot.
3. Remove the four bolts holding the robot to the skid.
4. Use properly rated equipment to move the robot to its new location.

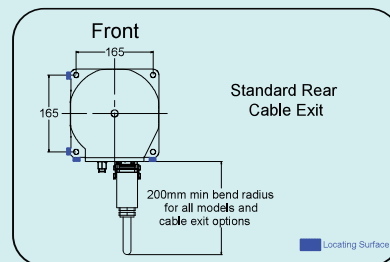
WARNING

Before moving heavy equipment, check and tighten any loose bolts. Do not pull eyebolts sideways. Otherwise, you will injure personnel or damage equipment.



5. Use (4) M10 bolts with washers to bolt the robot to a rigid platform or table that will prevent any motion of the robot base. See the diagram below for the ARC Mate 50iD/7L. Refer to the robot model-specific Operator's Manual for mounting pattern information.

NOTE: Be careful to avoid interference with bolts or connections.



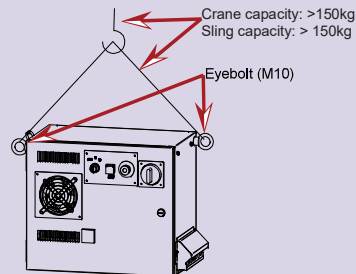
6. Secure the robot to the torque specified below. Remove the optional transportation brackets, if present.

Carbon Steel Grade Bolts M10 N·m : Newton Meters	
Socket Head Cap Screws 12.9	Hex Head Screws 10.9
64 N·m	64 N·m

Robot installation is complete. You are now ready to go to **B. Unpack and Install your Controller**.

B. Unpack and Install your Controller

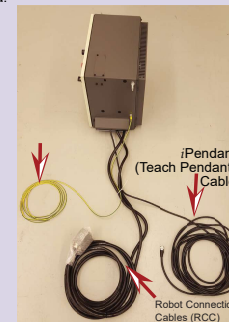
1. Use properly rated equipment to remove the controller from its box.



WARNING

Before moving heavy equipment, check and tighten any loose bolts. Do not pull eyebolts sideways. Otherwise, you will injure personnel or damage equipment.

2. Place the controller in a safe location where the grounding wire and RCC can reach the robot, and the controller is safely reachable from outside of the robot work area.



3. Connect the ground wire from the controller to the robot base. The grounding location has unpainted/bare metal as shown below for rear cable exit.

Grounding Location



Example Locator Pins

4. Connect the RCC from the controller to the base of the robot, and connect the teach pendant to its cable. Completed connections are shown to the right.



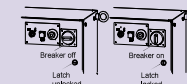
5. Connect the controller to power:

WARNING

Lethal voltage is present in the controller whenever it is connected to a power source. Be extremely careful to avoid electrical shock. Lock out and tag out the power source at the controller according to the policies of your plant.

- a. Find the power source needed for your controller on the door near the F Number. See diagram on the left side of this Guide for location. If a 120VAC side transformer is used, connect it to a 120VAC 20amp circuit.

- b. Open the controller door by using a straight-head screwdriver to turn the latch counter-clockwise, then turn the Breaker counter-clockwise past its off position.



Go to Page 2

MARGTAM5011161E REV C

- C. Route the power input cable through the side of the controller as shown in the diagram below.

R-30iB Mate Controller - Inside View



User-supplied
Power
input
cable

- d. To connect power, remove the plastic cover from above the Breaker (see the diagram below.) To remove the cover, insert a straight-head screwdriver into the opening, and push the tab to the left allowing the cover to slide forward.

Plastic
Breaker
cover
opening
location



Breaker
with
cover
removed

- e. Connect power to the Breaker with the customer-supplied power input cable. For Single Phase, connect L to 1 and N to 3. For Three Phase, connect L1 to 1, L2 to 3, and L3 to 5. Connect the ground to the plate to the right of the Breaker.



Breaker connection with
customer-supplied power input
cable

Breaker with
cover reinserted



- f. After the power cable is connected, reinsert the plastic cover.
- g. Use a multimeter to verify the correct voltage is present.

- h. Before you close the controller door, retrieve the set of keys from inside the door, and store them in a known location outside of the controller for use later in this guide.

Key location inside controller

- i. Close the controller door and use a straight-head screwdriver to turn the Latch on the outside of the controller counter-clockwise.



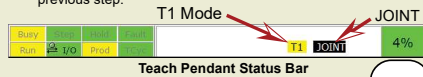
Controller installation is complete. You are now ready to go to **C Turn on Your Controller**.

C. Turn on your Controller

WARNING

Lethal voltage is present in the controller whenever it is connected to a power source. Be extremely careful to avoid electrical shock. Lock out and tag out the power source at the controller according to the policies of your plant.

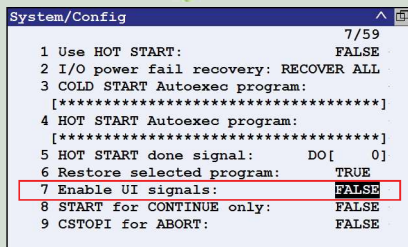
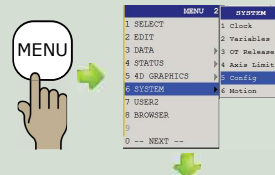
- Turn on the controller by rotating the Breaker clockwise to ON.
- Locate the Mode switch on the Controller Operator Panel. Use the key retrieved in section B, step 5h to switch to T1 mode. T1 mode limits the robot speed to 250mm/sec and requires robot control to be at the teach pendant only.
- Verify the controller is in T1 Mode by looking at the teach pendant screen. The mode indicator will be highlighted yellow and look similar to the status bar below. If it is not T1, repeat the previous step.
- Press the COORD key on the teach pendant until JOINT is displayed similar to the status bar above.



You have turned on your controller. You are now ready to go to **D. Jog your Robot**.

D. Jog your robot

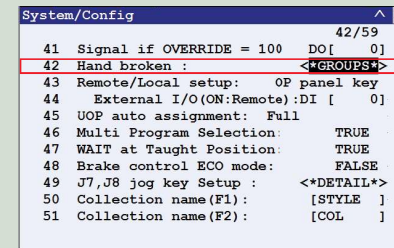
- To jog the robot in JOINT mode:
 - Turn the Teach Pendant ON/OFF switch to the ON position. Ensure the Emergency Stop buttons on the teach pendant and controller are popped out. If not, twist the button clockwise to reset.
 - Press MENU, select 0 (Next), and select (SYSTEM), and then select CONFIG.
 - Use the teach pendant arrow keys to move the cursor to the Enable UI signals option.
 - Use the teach pendant F5 key to set the option to FALSE.



- C. Use the teach pendant arrow keys to move the cursor to the Hand Broken option (around line 42) and press ENTER.

Hint: You can hold the SHIFT key and press the down arrow key to cursor through the list faster.

- Use the teach pendant F5 key to disable the hand broken error for each motion group.



- d. Press and hold a DEADMAN switch on the back of the teach pendant to its middle position. Press RESET on the teach pendant to reset alarms.

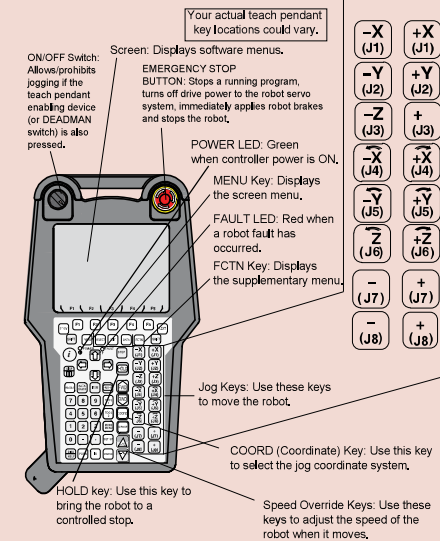
NOTE: To view alarms, press MENU, cursor to ALARM, and press ENTER. To toggle between alarm history and active alarms, press F3.

- e. To move each joint axis of the robot (called "jogging the robot in JOINT mode"), press and hold SHIFT and press each jog key one at a time and watch the robot joint move. Refer to the **Robot (mechanical unit) with joints indicated** diagram on Page 1 of this guide.

Try jogging each joint of the robot.

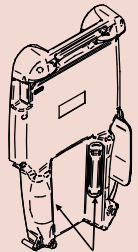
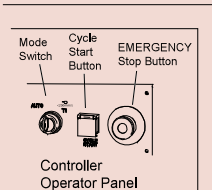
Jogging the robot is complete. You are now ready to perform software installation, if necessary. Refer to the *Controller-specific Software Installation manual*.

Reference



Refer to your *Application-Specific Setup and Operations Manual* for more information on each teach pendant key.

iPendant
(Teach Pendant)



For more information

To access controller, robot, or application-specific manuals, scan the QR Code to the right or go to <https://www.fanucamerica.com/>.



United States: 1-888-326-8287
Canada: 1-905-812-2300
Mexico: + 52 (449) 922-8000
South America: + 55 (11) 3619-0599